Week 2 Cybersecurity Project Report

Footprinting and Network Scanning

Report date: 10 November 2026 | Evidence: eight redacted screenshots

Executive Summary

This report records the Week 2 footprinting exercises and a network scan shown in the supplied evidence. The public-domain checks returned DNS, web technology, HTTP response, and web application firewall observations for networkwalks.com. A separate Zenmap scan of the VirtualBox lab subnet showed a responding host with TCP ports 135 and 445 open and 998 TCP ports filtered. No exploitation, credential testing, or proof of a vulnerability is shown in the evidence.

The screenshots include opaque redactions over IP addresses, MAC and contact details, an email address, and a session cookie. The original evidence files remain unchanged.

1. Liability Disclaimer

This educational report documents tool output visible in the supplied screenshots. Testing of systems should be performed only within a lab or with explicit written authorization from the system owner. The evidence identifies a VirtualBox lab scan and separate public-domain lookups for networkwalks.com; it does not include a signed authorization for testing that public domain. Accordingly, this report records the displayed observations and does not represent that authorization was obtained, that the domain is owned by the student, or that a security assessment of its owner was approved.

Results are limited to the captured tool output and the time of collection. Fingerprinting can be inaccurate or incomplete. Redacted screenshots should be used for sharing; the unmodified originals should be stored securely.

2. Introduction

The Week 2 project practiced reconnaissance and network-scanning workflows. The objectives were to collect registration and DNS information, identify web technologies and perimeter behavior, inspect HTTP response headers, and interpret a scan of the local lab network.

The report separates public-domain footprinting observations from the VirtualBox lab scan so that evidence from one environment is not attributed to the other. Findings below describe what the tools displayed; they are not independently validated security conclusions.

3. Tools Used

Tool	Purpose in this project
WHOIS	Reviewed public domain registration and name-server data.
WhatWeb	Fingerprinting of the web application and exposed response characteristics.
NSLookup	Lookups for A, NS, and MX records using the resolver shown in the screenshot.
cURL	Retrieved the HTTP response status and visible response headers.
WAFW00F	Checked whether the target appeared to be behind a web application firewall.
DNSRecon	Standard DNS enumeration; the output reported eight records.
Zenmap / Nmap	Scanned the VirtualBox lab subnet and displayed host, port, and service information.

4. Activities Performed

- Queried WHOIS registration data and reviewed registrar, status, and name-server fields.
- Used NSLookup and DNSRecon to inspect A, NS, MX, SOA, TXT, and SRV information displayed for networkwalks.com.

- Ran WhatWeb and WAFW00F to collect web technology fingerprints and a WAF identification result.
- Used cURL to capture the HTTP response status and headers, including cookie attributes; the cookie value is redacted in the report evidence.
- Used Zenmap with an intense scan profile against the VirtualBox lab subnet shown in the screenshot, then reviewed the host and service output.
- Prepared redacted copies of all eight screenshots for inclusion as report evidence.

5. Results and Findings

5.1 Public-domain footprinting observations

- WHOIS displayed GoDaddy as registrar, HostGator name servers, and an unsigned DNSSEC status. Registrar abuse contact details are redacted.
- NSLookup displayed an A record for networkwalks.com, two HostGator name servers, and an MX result pointing to mail.networkwalks.com. Resolver and returned IP addresses are redacted in the evidence.
- DNSRecon reported eight records, including SOA, NS, A, MX, TXT, and SRV data. Address values and a verification token are obscured in the screenshot.
- WhatWeb reported HTTP 200 and fingerprinted Apache, WordPress 7.1.4, jQuery 3.7.1, and Bootstrap 7.1.4. These are tool fingerprints, not independently verified component versions. The displayed contact email and IP are redacted.
- WAFW00F reported that the site was behind ModSecurity (SpiderLabs). This is a tool identification result and does not establish the effectiveness of the control.
- cURL displayed HTTP/2 200, an Apache server header, WordPress-related headers, and a Set-Cookie response with Secure and HttpOnly attributes. The session cookie value is redacted.

5.2 VirtualBox lab scan observations

- Zenmap scanned the 10.0.0.2/24 target range shown in the screenshot and reported the host 10.0.0.1 as up.
- The scan reported 135/tcp open (msrpc) and 445/tcp open (microsoft-ds), with 998 TCP ports filtered. The MAC address is redacted.
- The output showed a one-hop network distance and a Windows service/OS identification. OS guesses were explicitly uncertain, so they should be treated as tentative.

5.3 Interpretation

The supplied evidence supports a record of reconnaissance outputs and an exposed-service observation in the lab. It does not show successful exploitation, an access-control bypass, data exposure, or a confirmed vulnerability. No severity rating is assigned because the screenshots do not demonstrate impact or validate a weakness.

6. Recommendations

1. For the lab host, confirm that TCP 135 and 445 are required. Restrict them to trusted lab segments and disable services that are not needed; verify the result with an authorized rescan.
2. Keep the lab operating system and SMB/RPC services patched. Review SMB signing and protocol configuration on the host before drawing conclusions from the fingerprint alone.
3. For the owner of networkwalks.com, verify the reported software versions and update WordPress, themes, and plugins through the normal change process. The WhatWeb versions are fingerprints and may be inaccurate.
4. Review DNS record ownership and mail-sending needs. Confirm SPF, MX, and name-server values, and consider DNSSEC only after checking registrar and DNS-host support.
5. Review the complete HTTPS configuration and response-security headers with the domain owner’s approval. The captured cURL output is not sufficient to establish that any header is missing or that TLS is misconfigured.
6. Continue protecting evidence before sharing. Keep IP addresses, MAC addresses, contact details, email addresses, and session tokens masked in report copies.

7. Conclusion

The Week 2 exercises captured domain-registration, DNS, web-fingerprinting, WAF, HTTP-header, and lab-network scan evidence. The lab scan showed two open TCP services and a large filtered-port count on the responding host. The public-domain tools returned basic infrastructure and technology observations, but those results alone do not establish a vulnerability. The redacted screenshots preserve the main tool output while hiding the identified sensitive values.

8. Evidence Collected

The following redacted screenshots are included. Each is a copy of the supplied image; the source screenshots were left unchanged. Opaque masks cover the identified network and contact details while retaining surrounding tool output where possible.

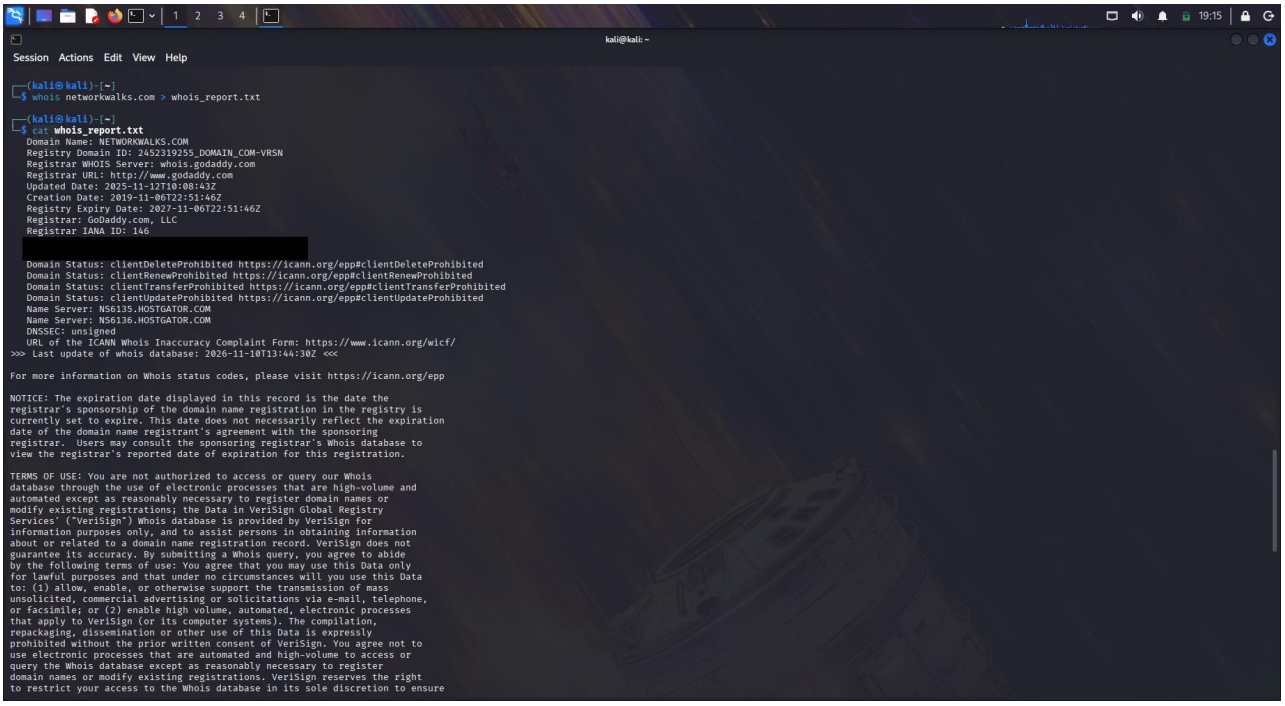


Figure 1. WHOIS registration output with abuse contact email and telephone redacted.

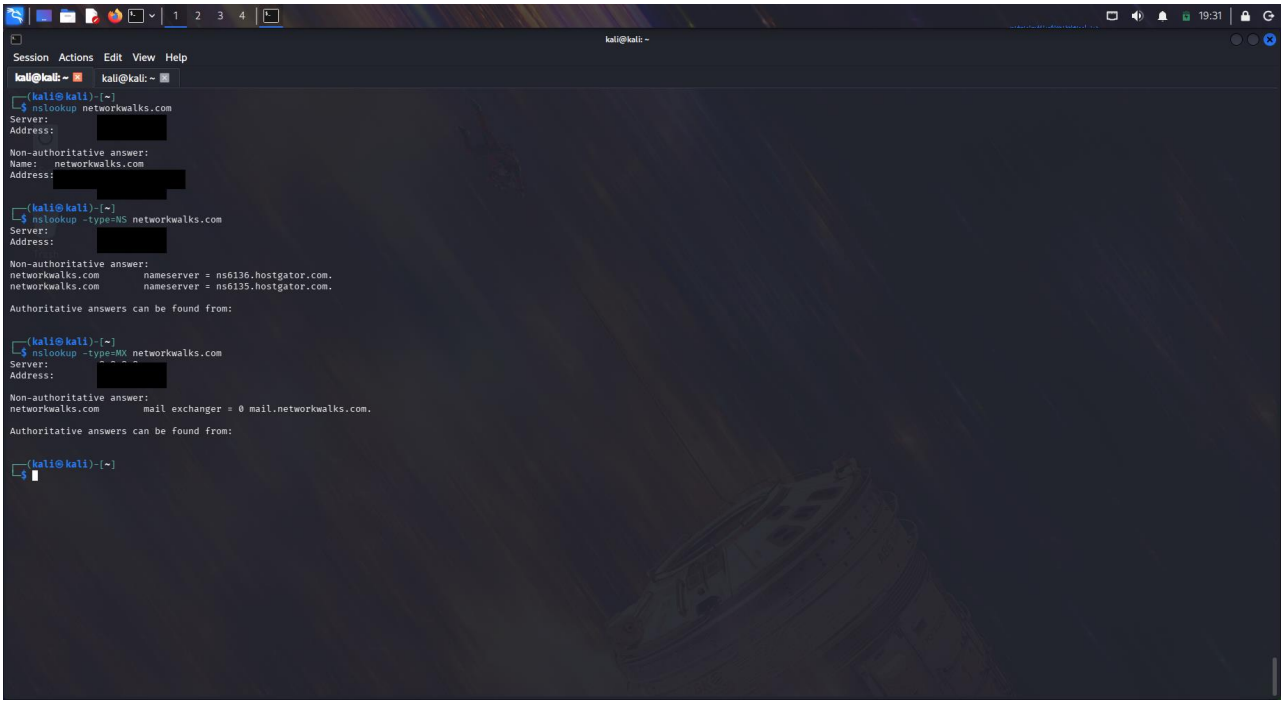


Figure 2. NSLookup A, NS, and MX results with IP addresses redacted.

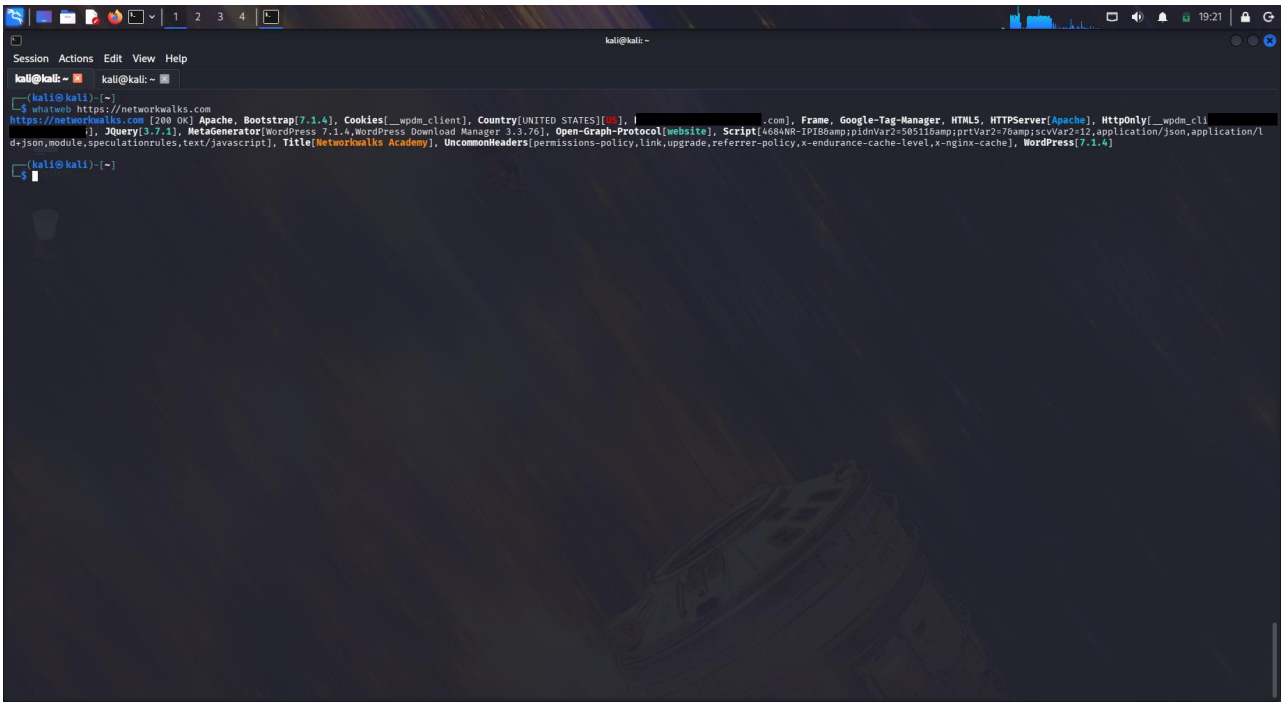


Figure 3. WhatWeb fingerprint with email address and IP address redacted.

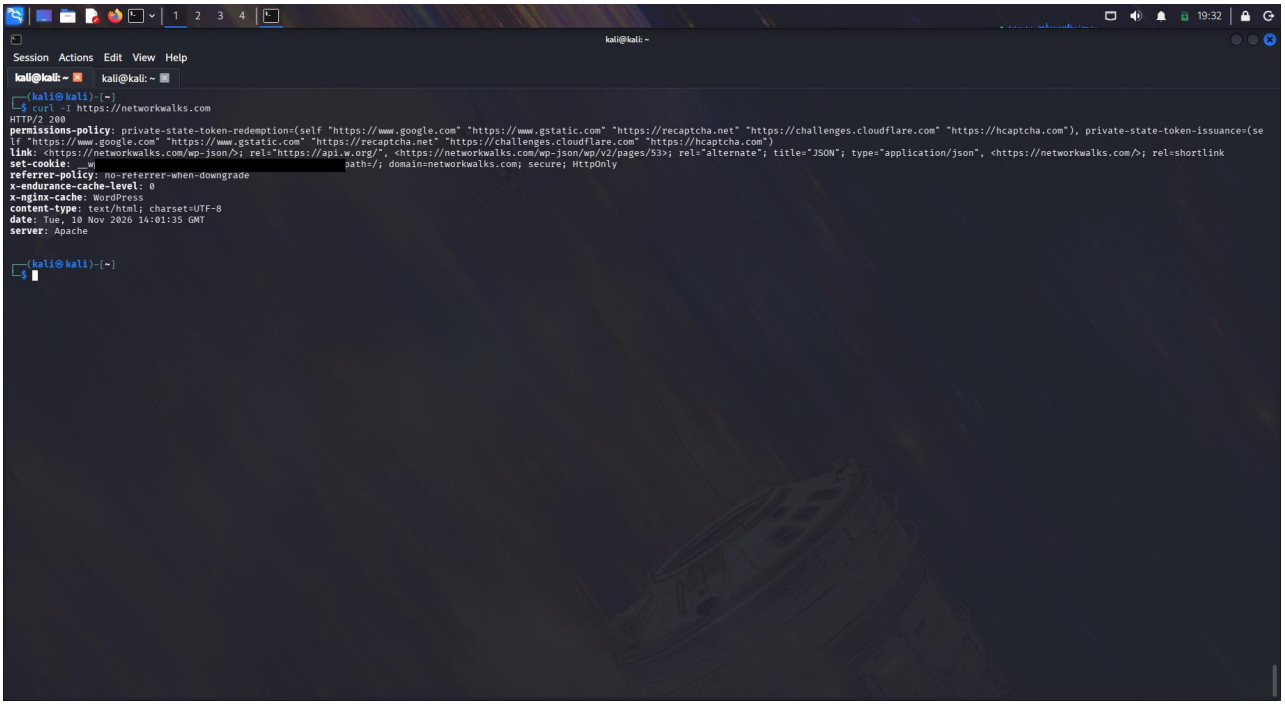


Figure 4. cURL HTTP response headers with the session cookie value redacted.

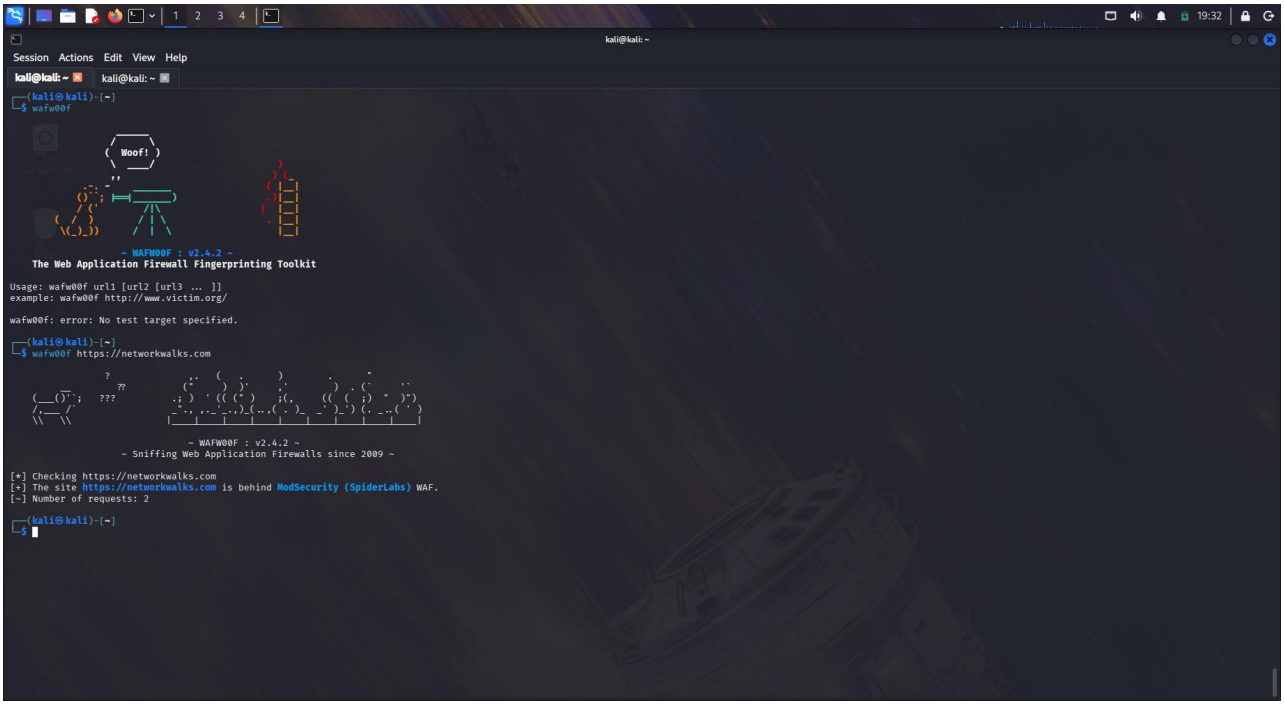


Figure 5. WAFW00F output identifying ModSecurity (SpiderLabs).

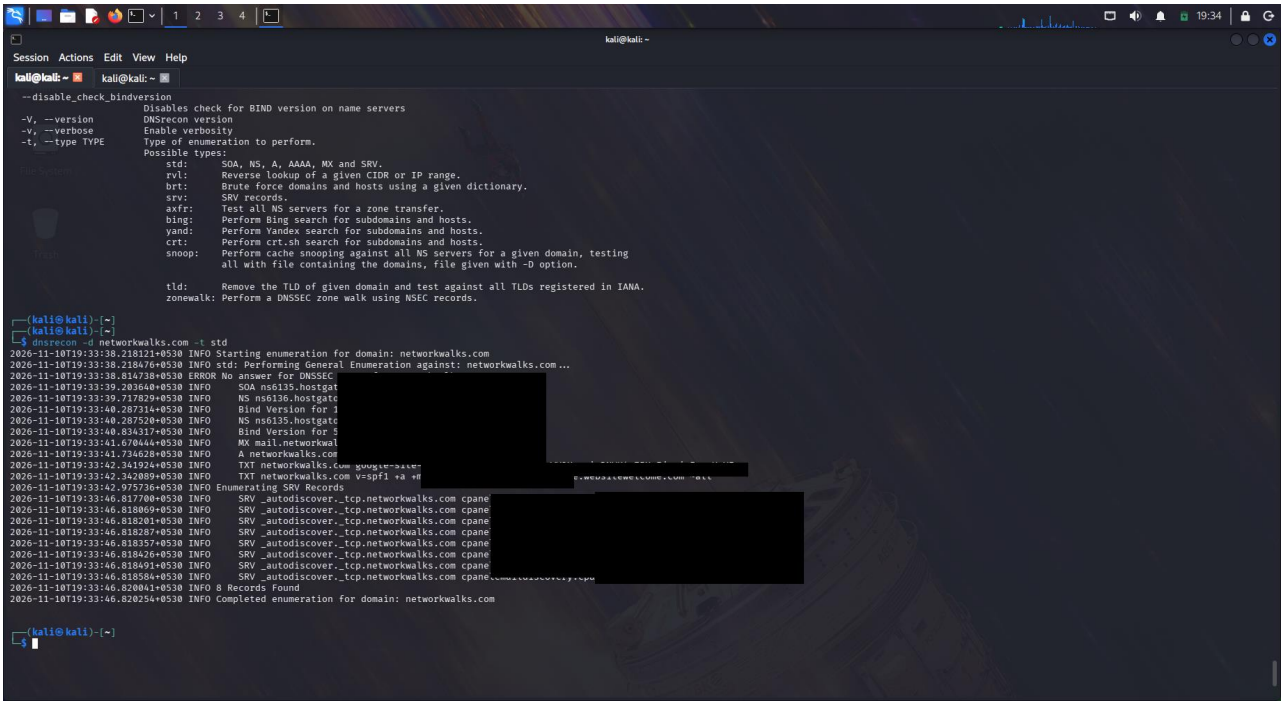


Figure 6. DNSRecon standard enumeration with returned address values and verification token masked.

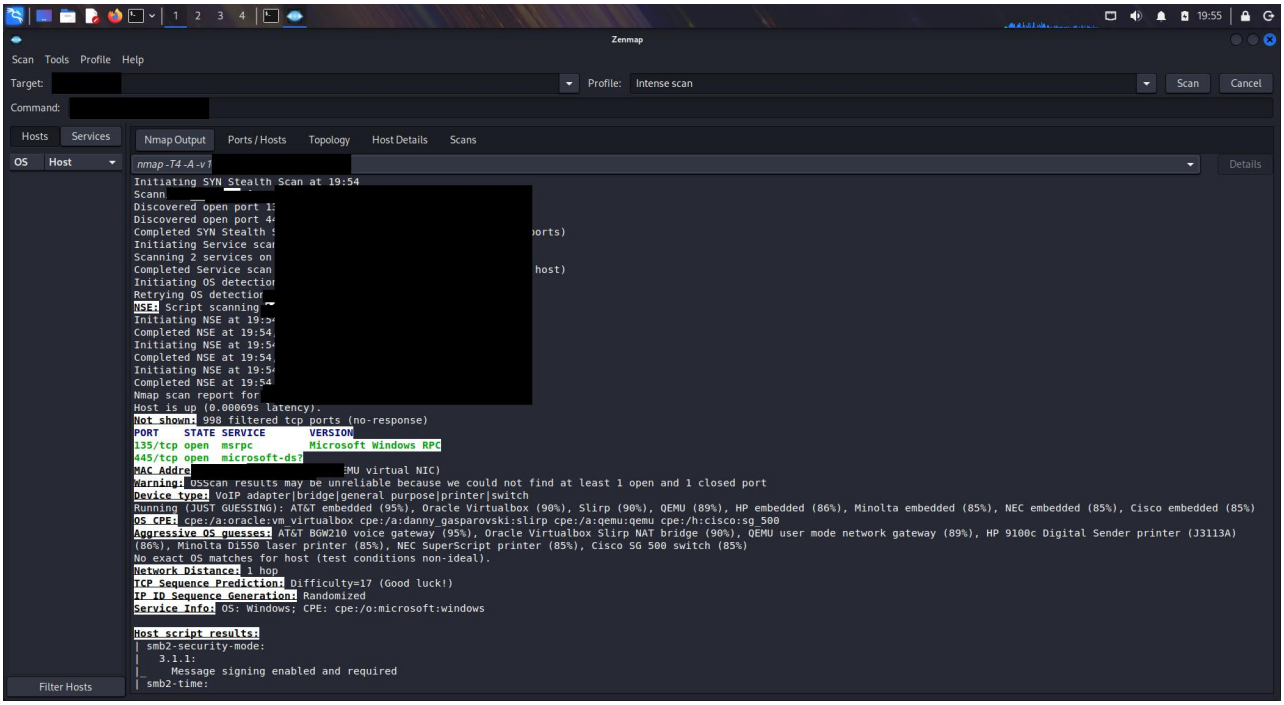


Figure 7. Zenmap Nmap output with target and host identifiers and MAC address masked.

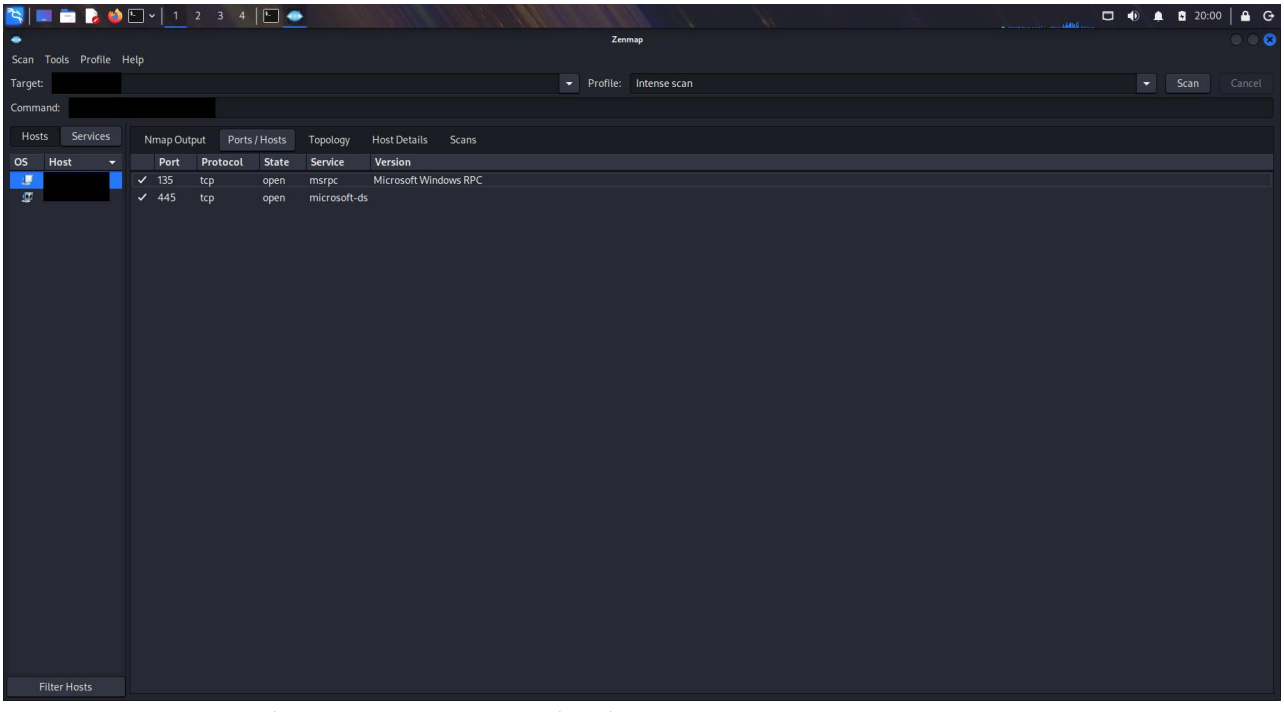


Figure 8. Zenmap Ports / Hosts view with target subnet and host addresses masked.